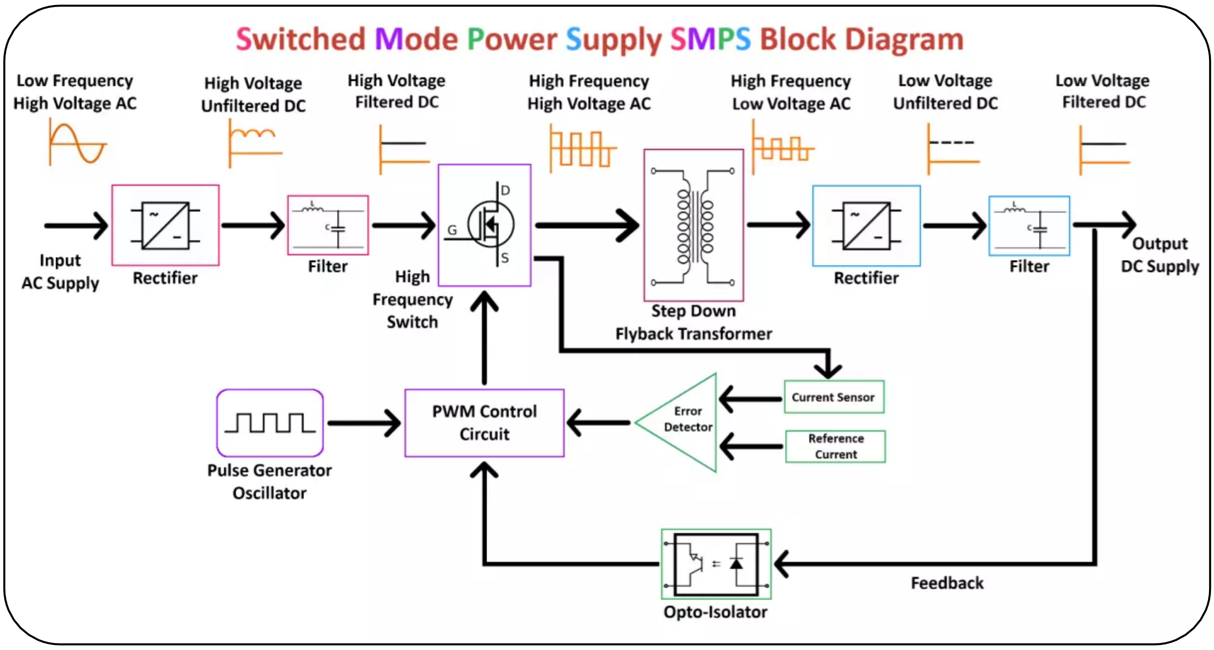
1. **List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.**

Devices using an SMPS (Switched Mode Power Supply)

- **Smartphone Chargers:** Converts the AC wall outlet into a safe, low-voltage DC level (e.g., 5V or higher) specifically tuned to charge your phone's battery rapidly and safely.
- **Laptops:** Steps down mains AC into specific DC voltages required to power motherboard components and charge the internal battery without adding bulky transformers.
- **Desktop Computers:** Delivers multiple regulated, highly-efficient DC outputs (like +12V, +5V, and +3.3V) required to run the CPU, graphics card, and storage drives.

2. **Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.**



3. **Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each,**

write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-Pin Motherboard Connector

- **Devices Powered**: Delivers the primary electrical power to the entire motherboard, distributing power to the chipset, memory slots, and integrated components.
- **Safety Tip**: Depress the plastic retaining latch completely before pulling, and gently rock the connector side-to-side rather than yanking it straight out to avoid ripping the socket off the motherboard layers.

2. SATA Power Connector

- **Devices Powered**: Supplies power to internal storage drives like Solid State Drives (SSDs), Hard Disk Drives (HDDs), and optical disc drives.
- **Safety Tip**: Ensure the distinctive L-shaped guide key is aligned correctly before inserting to prevent breaking the thin plastic structural housing on the drive's data/power interface.

3. Molex Connector (4-Pin Peripheral)

- **Devices Powered**: Powers legacy devices, extra case fans, water cooling pumps, and various internal desktop lighting components.
- **Safety Tip**: Inspect the four metal pins inside the plastic housing before connecting; they frequently become loose or misaligned, which can bend the internal metal or cause a severe short circuit if forced.

4. PCIe 6/8-Pin Connector

- **Devices Powered**: Provides high-wattage auxiliary power to dedicated PCI Express graphics cards (GPUs).
- **Safety Tip**: Avoid confusing the 8-pin PCIe (6+2 configuration) cable with the 8-pin EPS/CPU (4+4 configuration) cable, as forcing the wrong electrical pinout into a slot will permanently fry your hardware.

4. **Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.**

➢ Here is the step-by-step guide to connecting your Switched-Mode Power Supply (SMPS) to your gaming PC components.

1. Mount the SMPS

- **Action**: Place the SMPS into the PC case power supply shroud.
- **Detail**: Ensure the fan faces the ventilation cutout (usually downward).
- **Security**: Secure it tightly to the back of the case using four PSU screws.

2. Connect the Motherboard Power

- **Cable Used**: Main **24-pin ATX** cable.
- **Action**: Plug the large 24-pin connector into the matching long slot on the right side of the motherboard.
- **Detail**: Push until the plastic latch clicks firmly into place.

3. Connect the CPU Power

- **Cable Used**: **8-pin (4+4) EPS** cable.
- **Action**: Route this cable to the top-left corner of the motherboard.

- **Detail**: Split or combine the 4+4 pin connector to match your motherboard's CPU power slot (some require 4, 8, or 12 pins total).

4. Connect the Graphics Card (GPU)

- **Cable Used**: **6+2 pin PCIe** cable (or a **12VHPWR** cable for newer NVIDIA cards).
- **Action**: Plug the required number of pins into the top edge of the graphics card.
- **Detail**: Use separate, dedicated PCIe cables from the SMPS for high-end GPUs rather than daisy-chaining a single cable.

5. Connect the Storage Devices

- **Cable Used**: **SATA Power** cable (flat, wide connector).
- **Action**: Plug the L-shaped SATA power connector into your 2.5-inch SSDs or 3.5-inch HDDs.
- **Detail**: Note that M.2 NVMe SSDs install directly onto the motherboard and do not require SMPS cables.

To ensure a safe first boot, let me know:

- The **exact model of your graphics card** (e.g., RTX 4070, RX 7800 XT)
- The **wattage and model of your SMPS**

I can verify if you need a specific modern adapter or confirm your total power headroom.